

Noise Floor Analysis

Test Report

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Report No: SER0005

Document Classification: External

Preface

This document presents the findings and analysis of the Noise Floor present in Jaisalmer site.

Privacy Information

This document may contain information of a sensitive nature. This information should not be given to any persons outside of the Avartanam test labs LLP. without prior consent.

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1.General Information

Service Request No.	SER0005-0324
Purchase Order No/Quotation No	
Name and Address of Customer	Mr.Diptesh Patel System Level Solutions (India) Pvt Ltd. Plot No. 32,Zone D-4,Phase 1,G.I.D.C. Estate., Vithal Ugyognagar - 388121, Anand - Gujarat - India.

2. Introduction

- This document describes the information about Noise Floor Analysis performed at various sites in Jaisalmer and Importance of Avartanam (ATL) Filter.

2.1 Why Noise Floor test Performed?

- To identify presence of unwanted signal (Noise signal) in our operating frequency band as well as nearby out of frequency band.

3 DIV Site

- In DIV Site, Cellular tower installed very nearer to Repeater and observed that Noise level in out of band obtained quite higher compared to Actual our signal strength.

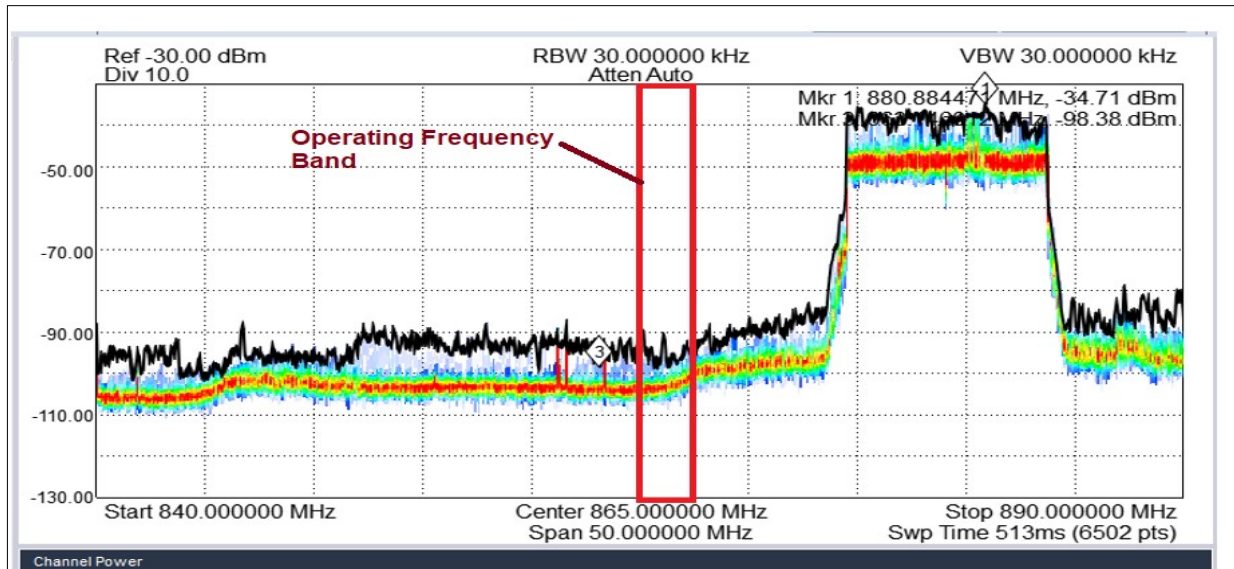


Figure 1: Noise Floor without ATL Filter

- PFA the images which showcase Noise present without Filter in Fig.1 & with filter in Fig.2

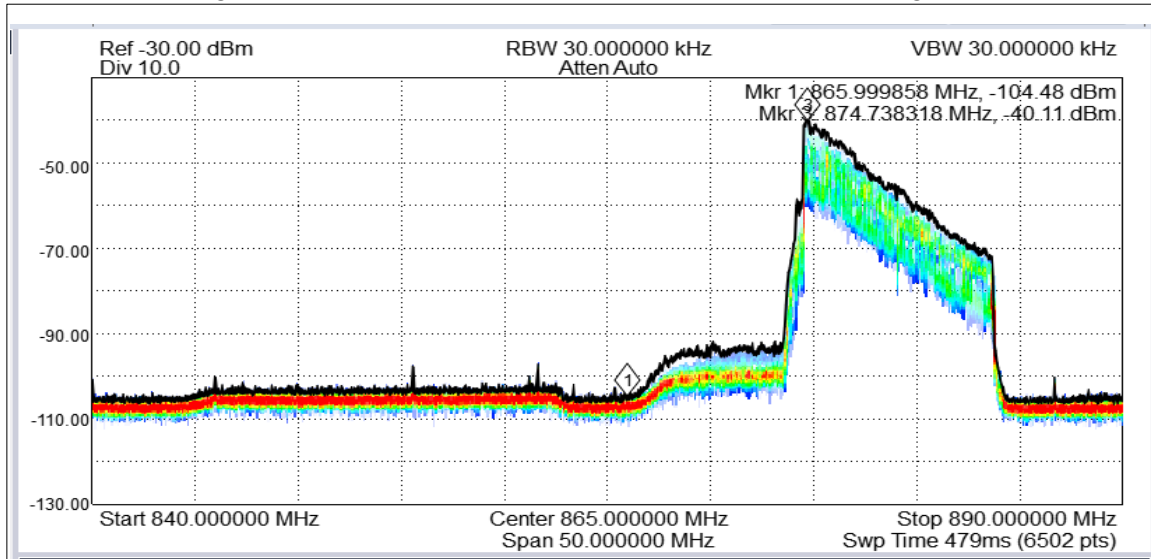


Figure 2: Noise Floor with ATL Filter

- As seen above plot in Fig.2 , Noise level heavily reduced with filter.

4 Chain Singh Site

- In chain Singh site also observed that cellular tower installed 30 to 40 m apart from DCU.

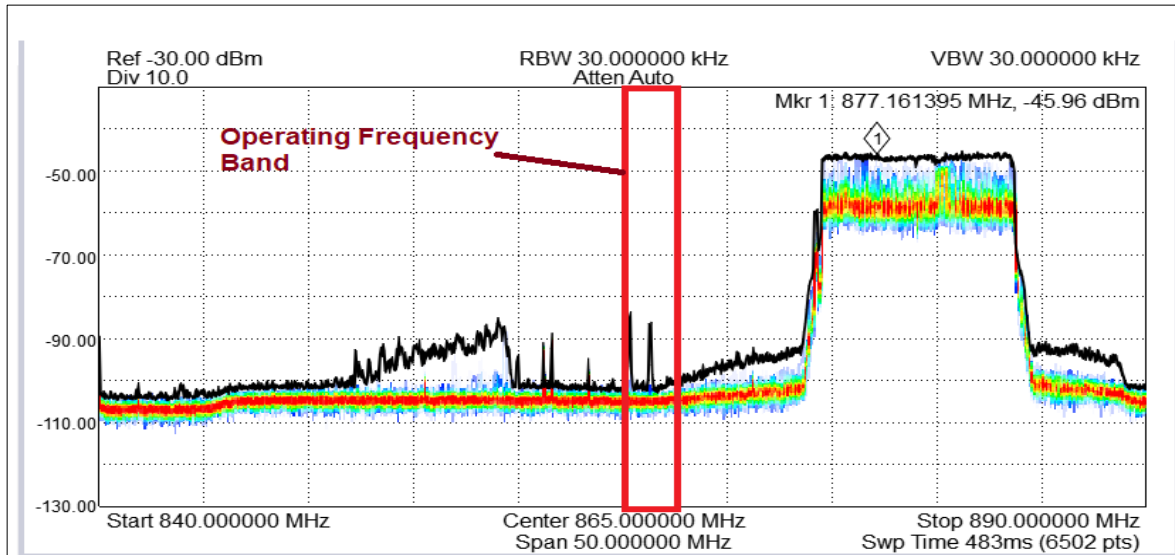


Figure 3: Noise Floor without ATL Filter

- As shown above plot in Fig.3, Observed that Noise level for out of band obtained higher due to cellular tower.
- Noise level sufficiently reduced with ATL (Avartanam)Filter and established communication smoothly as seen below Fig.4

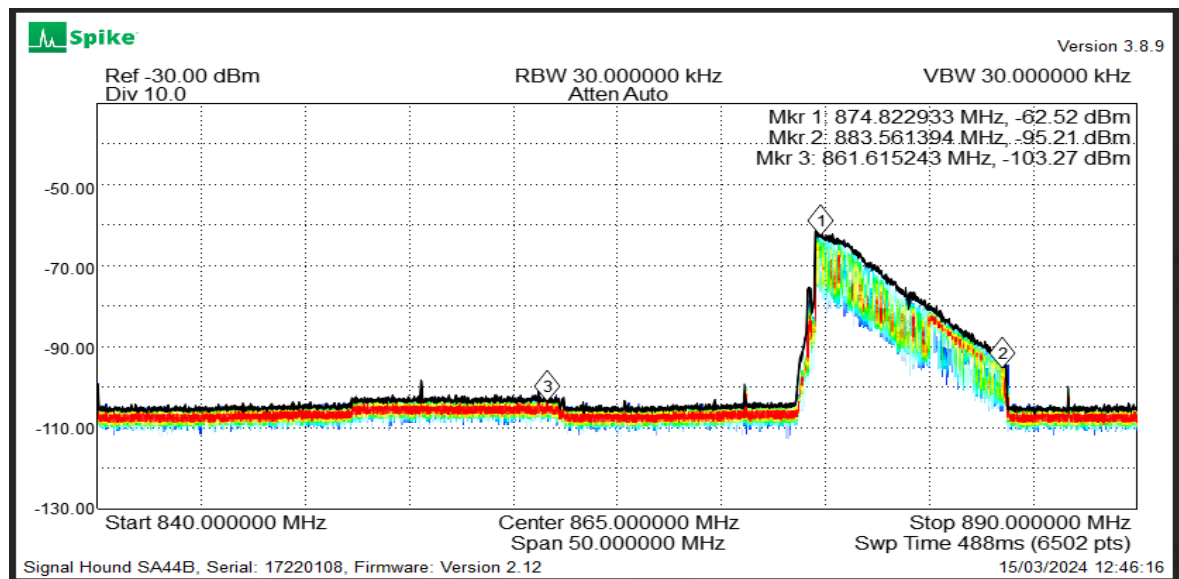
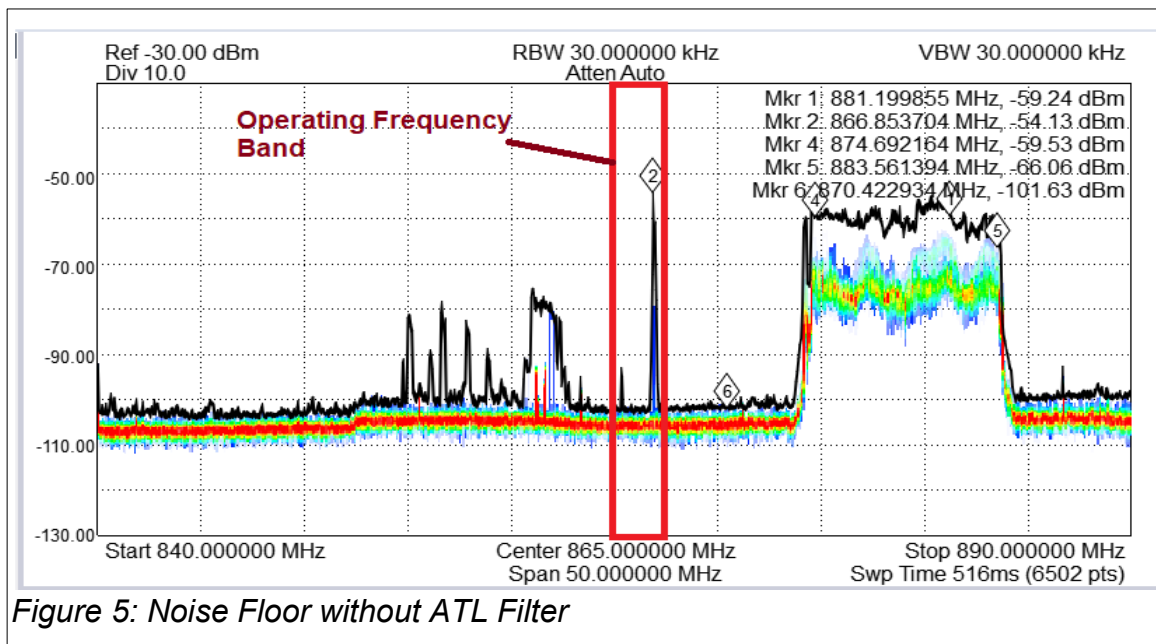


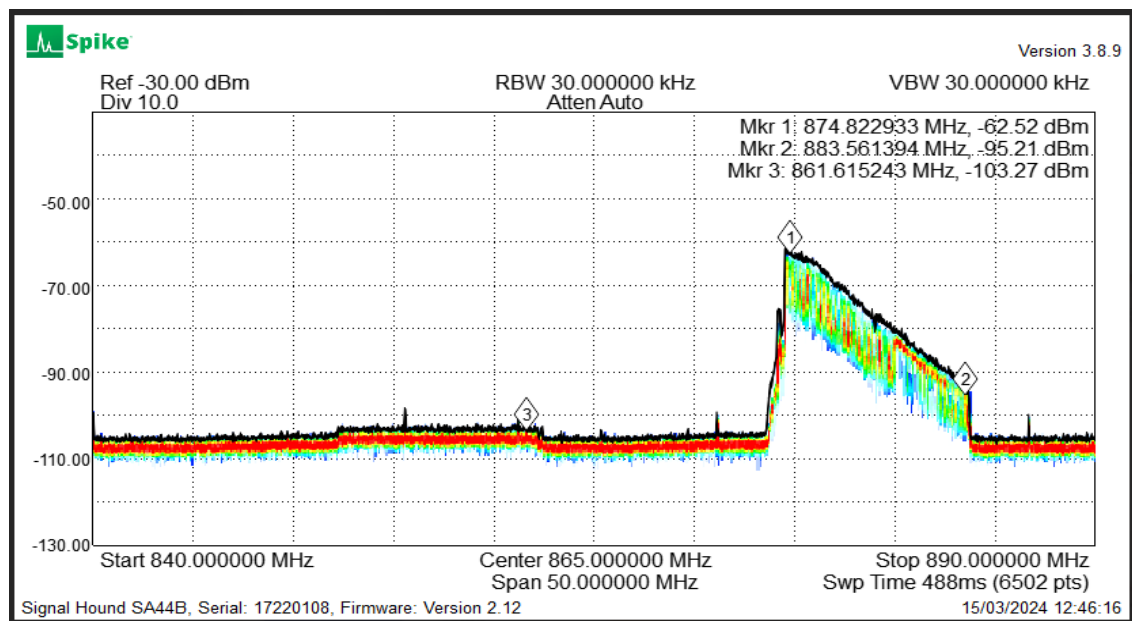
Figure 4: Noise Floor with ATL Filter

5 SONAR Site

- In SONAR Site, Cellular Tower installed 200 m apart from new DCU. Although Noise level observed quite higher in nearby out of frequency band as seen Fig. 5



- Please find below plot with ATL filter in Fig.6



6 Mayur Vihar Site

- In Mayur Vihar Site, Observed that Noise Floor level is Higher due to Cellular Tower .

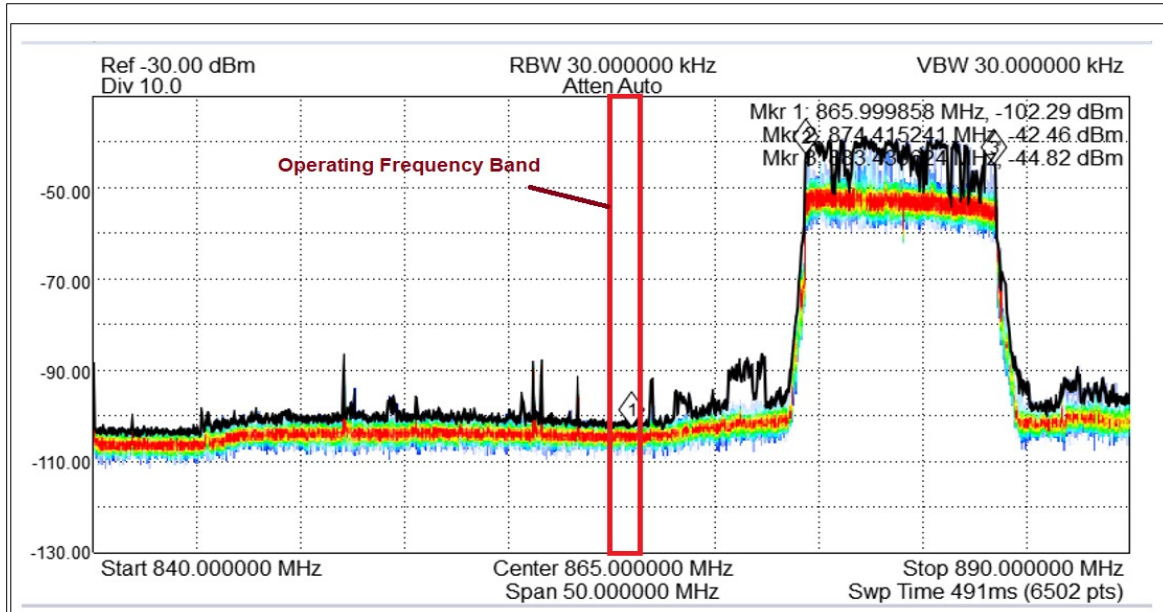


Figure 7: Noise Floor without ATL Filter

- Please find below plot with ATL filter in Fig.8

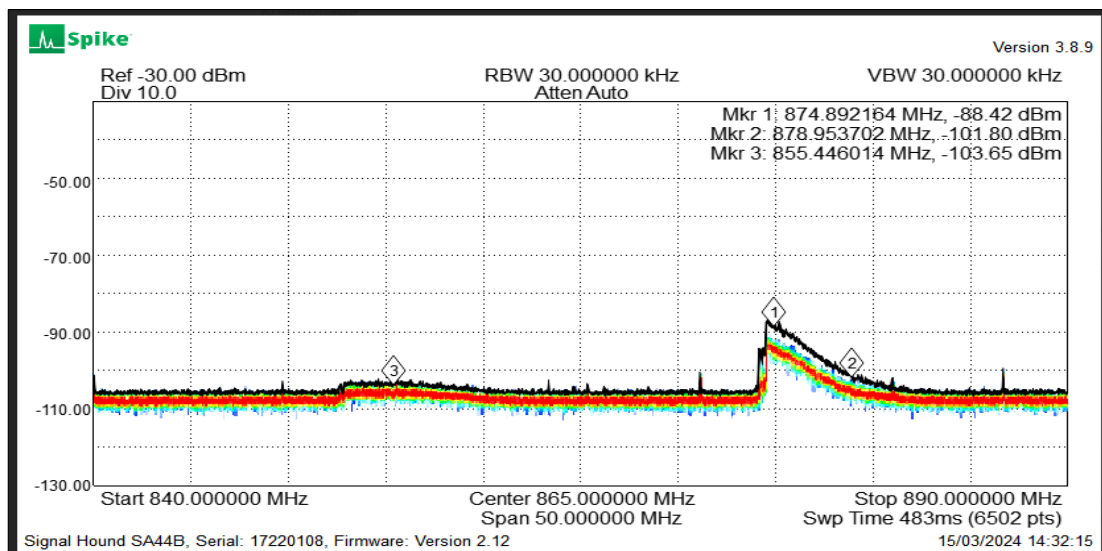


Figure 8: Noise Floor with ATL Filter

7 Khambata Site

- In Khambata Site, Cellular tower installed approximately 100 meter from DCU.

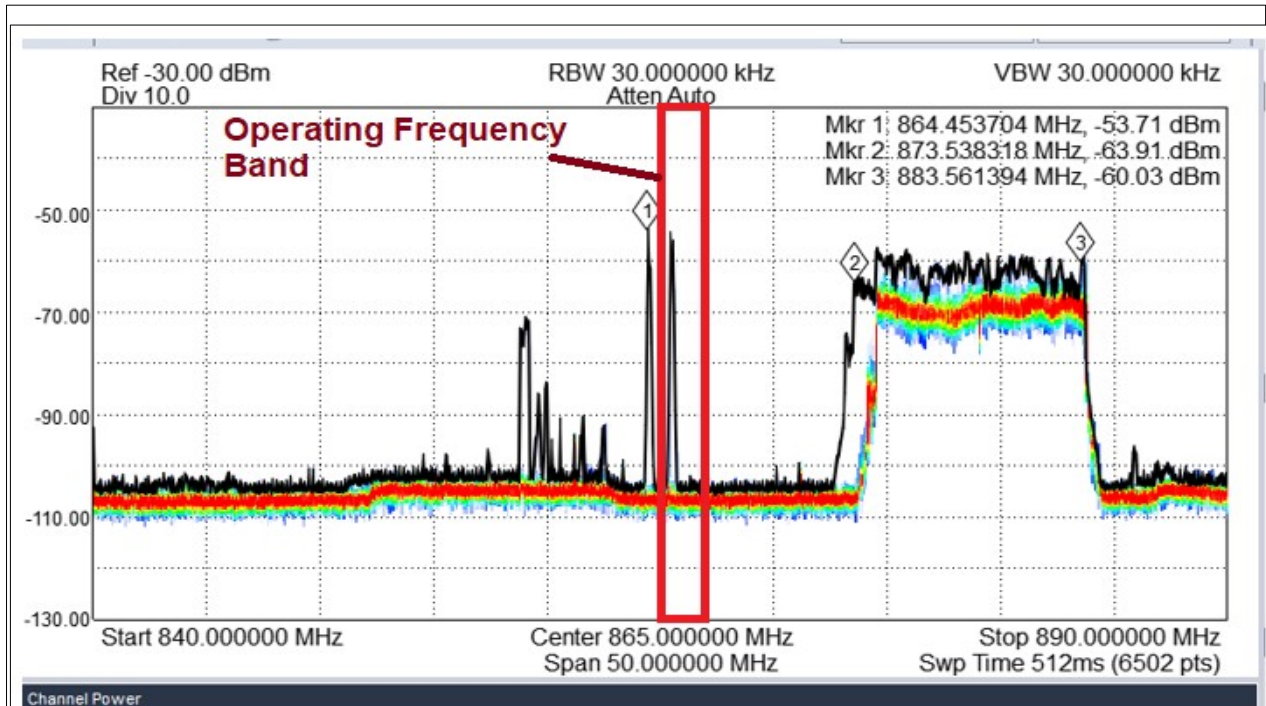


Figure 9: Noise Floor without ATL Filter

- Please find below Noise Floor plot with ATL Filter

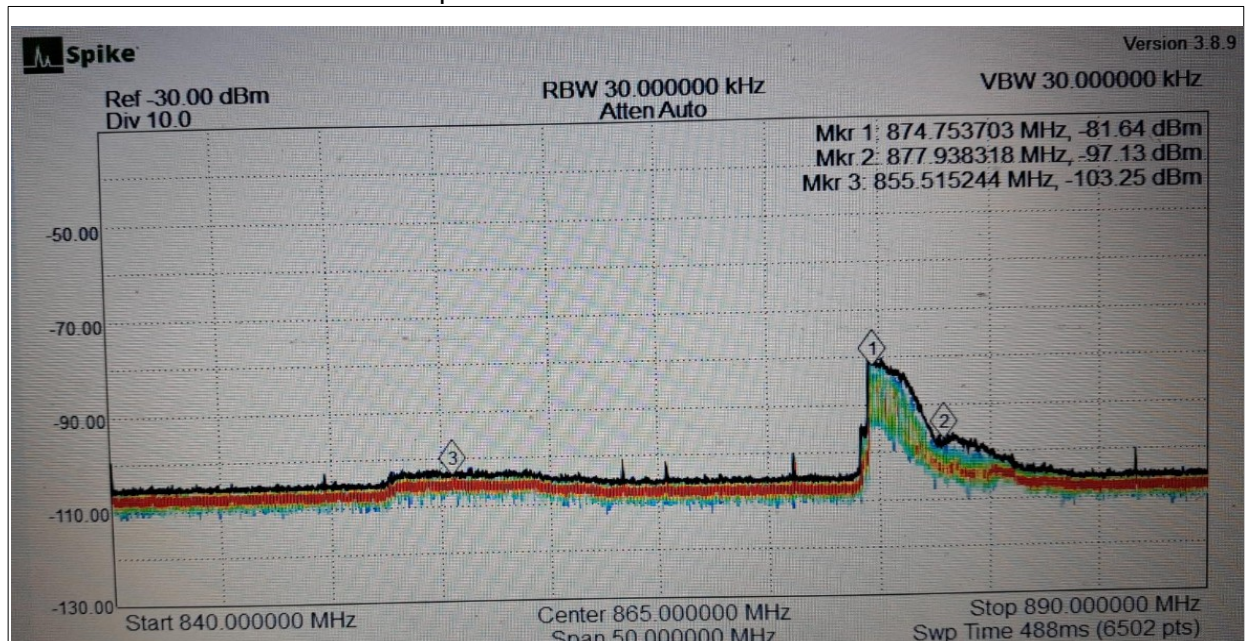


Figure 10: Noise Floor with ATL Filter

8 Observation

- Regarding Jaisalmer Site Survey, Avartanam Filter solving below three major hurdles of Wireless communications.
 - **Out of Band Interference:** If the wireless device operates on frequencies close to those used by nearby cellular towers, the signals from these towers can interfere with the device's signals. This interference can disrupt communication, reduce signal strength, and degrade data transmission rates.
 - **Spectrum Congestion:** Here, multiple cellular towers may operate on overlapping frequency bands, leading to spectrum congestion. This congestion can increase interference and competition for available bandwidth, further degrading the quality of wireless signals.
 - **Low Noise Amplifier :** Low Noise Amplifier which are main block of wireless communication to make communication stable, is saturated due to this heavy Out of Band Noise and its Gain is degraded. Avartanam (ATL) Filter is suppressing this noise as seen above plots and prevent LNA to go under saturation.